

Evaluating the performance of object
Detection models? for Accurate Sri Lankan
Traffic Sign Detection

PROJECT DIARY

BCAS – SCHOOL OF COMPUTING
DEPARTMENT OF COMPUTING
BTEC HND PROGRAMME

Centre Ref. No:

Edexcel No:

Student Name: MN. SIHANAS

Supervisor Name: Mrs. NF. Mafaza.

Batch: HND in Computing - 22

INSTRUCTIONS FOR MAINTAINING THE PROJECT PROGRESS LOG

1. As credit will be given for the maintenance of the Project progress log all you interim and final assessment you are advised to maintain it neat and tidy and to keep it safe till the end of your Project submission period.
2. It is important that entries should be made at the end of each day's work and weekly entries at the end of each working week. All entries should be in ink.
3. Daily entries should indicate as clearly as possible about various work performed by you each day in your Project
4. The weekly entries should contain a brief description of the work done each week and about the issues analyze while meeting the Project supervisor
5. In the blank space provided write any new information you may have gathered as reference that may be necessary relevant notes from manufacturer's specification sketches, diagrams and technical drawings etc that will be useful to you. Give a brief description of the problems that you may have encountered, how they were overcome and steps taken to avoid their recurrence in the future
6. You are advised to make entries of any technical or managerial problems you may have encountered
7. You should obtain the signature of your Project supervisor each week on the day specified by him
8. The diary should be available for inspection whenever called for by Project coordinator or course coordinator.

Project Coordinator
British College of Applied Studies
School of Computing

Project Progress Log Report Page

PEARSON No				
Centre Ref. No				
Name	MN. Sihana s			
Project Working Period	From	2025-2-17	To	2025-8-15

Project Task Break Down Structure

Week	Project Task Completed	Starting Date	Ending Date
1	Preliminary research	02-17	02-23
2	Literature review	02-24	03-03
3	Dataset collection	03-04	03-10
4	Selecting models for evaluation	03-11	03-17
5	Reviewing Relevant papers to understand the problem	03-18	03-24
6	Literature review from case	3-25	3-31
7	Review models and datasets	4-01	4-7
8	Review and identifying performance metrics	4-8	4-14
9	Reviewing model enhancements	4-15	4-21
10	Literature review, data prep	4-22	4-28
11	Data preprocessing and cleanup	4-29	5-5
12	Preparing File structure for training	5-6	5-12
13	Performing data augmentation and normalizing	5-13	5-19
14	dataset sanity checking	5-20	5-26
15	dataset compressing & storing	5-27	6-2
16	yaml file creation & slide prep	6-3	6-9

upto
june
progress
should b
update

Week	Starting Date	Ending Date	Task Completed
17	6-10	6-16	progress presentation revision
18	6-17	6-23	training script creation
19	6-24	6-30	Preparing eval script
20	7-1	7-7	Comparing performance variation
21	7-8	7-14	Selecting best model based on performance
22	7-15	7-21	User interface req defining
23	7-22	7-28	Setting up dev environment
24	7-29	8-4	merging model with interface
25	8-5	8-11	final presentation prep.
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Project Progress Daily Sheet

Week Start Date	02 - 17
Week Ending Date	02 - 23

DAY	ACTIVITY
Monday	Preliminary research
Tuesday	
Wednesday	Selected research title.
Thursday	
Friday	Prepared proposal form.
Saturday	Prepared presentation slides. Proposal
Sunday	Submitted proposal form & Presentation Slides.

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

Prepared & submitted proposal form & slide.

Problems Occurred:

Suggestions Made by the Supervisor:

Provided suggestions to make changes on the proposal form.

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

005/02/23

DATE

N. Anand

PROJECT SUPERVISOR'S SIGNATURE

BCAS – Project & Development

Project Progress Daily Sheet

Week Start Date	02-24
Week Ending Date	03-03

DAY	ACTIVITY
Monday	
Tuesday	Literature Review.
Wednesday	
Thursday	
Friday	added F1-Score as a dependent variable.
Saturday	
Sunday	

Details of Work Carried out, Problem encountered how It is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever It is applicable).

Major Tasks Completed:

Literature Review.

Problems Occurred:

Suggestions Made by the Supervisor:

Suggested to add an extra dependent variable
+ ex) - FI - Score

Decisions Taken:

Added FI-score as a dependent variable.

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

2025/03/23
DATE


PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	03-08-4
Week Ending Date	03-10

DAY	ACTIVITY
Monday	
Tuesday	DataSet Collection
Wednesday	
Thursday	
Friday	
Saturday	
Sunday	

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

Dataset Collection

Problems Occurred:

Suggestions Made by the Supervisor:

Suggested to change the Dataset with a high amount of dataset.

Decisions Taken:

Dropped the old dataset & ~~not~~ collected a new dataset.

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

2025/03/23

DATE



PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	03 - 11
Week Ending Date	03 - 17

DAY	ACTIVITY
Monday	
Tuesday	Selecting the models for evaluation
Wednesday	
Thursday	Compared the dataset usage in literature. (article-1)
Friday	 Mirshu used traffic datasets like Lisa-TS, and GTSD for model evaluation.
Saturday	
Sunday	

Details of Work Carried out, Problem encountered how It is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever It is applicable).

Major Tasks Completed:

Literature review from
Article one (Mircha et.al.)

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

2025/04/28
DATE

Niraj P.
PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	03-18
Week Ending Date	03-24

DAY	ACTIVITY
Monday	Reviewing Relevant researches for understanding the methodologies.
Tuesday	
Wednesday	
Thursday	Reviewed YOLOv8 model advancements and performance from (Fathima et al) article-2
Friday	↓ Fathima fathima showed YOLOv8 achieved higher FPS but slightly less accurate than RTMDet,
Saturday	
Sunday	

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

Literature review from
Fathima et al.

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

2025/04/28
DATE

NATASHA
PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	03 - 28 25
Week Ending Date	03 - 31

DAY	ACTIVITY
Monday	
Tuesday	Understand small object detection improvements. - (Cao et al.)
Wednesday	
Thursday	
Friday	Summarized Analyzed detection challenges. from (Cao et al.)
Saturday	↓ Cao improved Sparse R-CNN with a Res2Nest backbone to better detect small traffic signs.
Sunday	

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

Literature review from
Cao et. al. (article-3)

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

2025/04/25
DATE


PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	04 - 01
Week Ending Date	04 - 07

DAY	ACTIVITY
Monday	Review models and datasets from Mircha et al.
Tuesday	Identified the evaluation metrics used in the previous researches.
Wednesday	Identified object detection models reviewed by Mircha et al. YOLOV3, SSD, Faster R-CNN, CenterNet etc.
Thursday	Also he used datasets below. Lisa-TS, GTSD, TTK100K, MTSD.
Friday	
Saturday	
Sunday	

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

Reviewed model/s and dataset from
Mirsha et al.

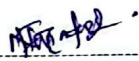
Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

2025/04/28
DATE


PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	04 - 08
Week Ending Date	04 - 14

DAY	ACTIVITY
Monday	Understand performance metrics and evaluation methods.
Tuesday	He used ^{compared}  He evaluated models using mAP@0.5, precision, and recall.
Wednesday	Mirshu et al. used mAP@0.5, compared FPS on GPU and grouped fine classes into 4 meta classes for traffic signs.
Thursday	
Friday	
Saturday	
Sunday	

Details of Work Carried out, Problem encountered how It is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever It is applicable).

Major Tasks Completed:

Reviewed and Identified performance evaluation metrics used by Mircha et al.

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

DATE

PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	04-15
Week Ending Date	04-21

DAY	ACTIVITY
Monday	
Tuesday	Reviewed models and advancements from Fathima et al. 
Wednesday	Fathima used (Fathima et al.) YOLOv8, YOLOv9, Faster-RCNN, RTMDet, Retina Net models. 
Thursday	Fathima et al. found out YOLOv8 has higher FPS but RetinaNet RTMDet was more balanced for road sign detection.
Friday	
Saturday	
Sunday	

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

Reviewed models & enhancements from Fathima et al.

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

05/05/2025
DATE


PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	04 - 22
Week Ending Date	04 - 28

DAY	ACTIVITY
Monday	
Tuesday	Review dataset preparation and evaluation metrics from Ruthim et al. ↓
Wednesday	Fathima used mAP and FPS for evaluating these models. ↓
Thursday	And Applied augmentations like noise and blur to enhance detection robustness. ↓
Friday	She used auto orientation and noise addition augmentations.
Saturday	
Sunday	

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

Understand metrics and data preparation
Used by Fathima et. al.

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

2025/04/28

DATE



PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	29-04 04-29
Week Ending Date	05-05 05-05

DAY	ACTIVITY
Monday	
Tuesday	Sort and filter dataset for training, valid, and testing.
Wednesday	
Thursday	Inspect the dataset for duplicate and faulty annotations.
Friday	Removing and correcting erroneous annotations in the dataset.
Saturday	
Sunday	

Details of Work Carried out, Problem encountered how It is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever It is applicable).

Major Tasks Completed:

Dataset pre-processing and cleanup.

Problems Occurred:

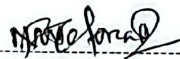
Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

05/06/2025

DATE



PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	05-06
Week Ending Date	05-12

DAY	ACTIVITY
Monday	Prepare label files and directories structure for training.
Tuesday	
Wednesday	Develop dataloaders for training pipeline.
Thursday	
Friday	
Saturday	
Sunday	

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

preparing File structure for training.

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

DATE

PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	05 - 13
Week Ending Date	05 - 19

DAY	ACTIVITY
Monday	Resizing necessary images.
Tuesday	
Wednesday	Performing data augmentation.
Thursday	
Friday	
Saturday	
Sunday	

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

performing data augmentation and resizing.

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

5/20/2025
DATE


PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	05 - 20
Week Ending Date	05 - 26

DAY	ACTIVITY
Monday	
Tuesday	Implement data transformation routines. → standardizing.
Wednesday	
Thursday	Enz performing initial data sanity check.
Friday	
Saturday	
Sunday	

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

data sanity check.

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

DATE

PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	05 - 27
Week Ending Date	06 - 02

DAY	ACTIVITY
Monday	
Tuesday	Progress presentation preparation
Wednesday	
Thursday	Zippping and uploading datasets across multiple cloud storages.
Friday	
Saturday	
Sunday	

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

dataset storing

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

03/06/2025

DATE



PROJECT SUPERVISOR'S SIGNATURE

BCAS – Project & Development

Project Progress Daily Sheet

Week Start Date	06 - 03
Week Ending Date	06 - 09

DAY	ACTIVITY
Monday	
Tuesday	Creating yaml files for detailing classes and settings.
Wednesday	
Thursday	Progress slide preparation.
Friday	
Saturday	
Sunday	

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

yaml file creation and slide preparation

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

DATE

PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	06-10
Week Ending Date	06-16

DAY	ACTIVITY
Monday	
Tuesday	Progress presentation slide review.
Wednesday	
Thursday	performing initial training small training to validate dataset functionality.
Friday	
Saturday	
Sunday	

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

Progress presentation review. &
- Small training process to test.

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

18/06/2023
DATE


PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	06 - 17
Week Ending Date	06 - 23

DAY	ACTIVITY
Monday	progress presentation already delegated. <i>work further</i>
Tuesday	
Wednesday	Develop training configuration. (epoches, batch size, LR)
Thursday	
Friday	Allocate computational resources.
Saturday	Preparing training Script.
Sunday	

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

Training Script creation. -

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

2025/6/30

DATE



PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	06-24
Week Ending Date	06-30

DAY	ACTIVITY
Monday	Implement custom callbacks. (model checkpoint, early stopping)
Tuesday	
Wednesday	Setup tools for training Visualization.
Thursday	
Friday	Prepare evaluation script. (calculate mAP, F1-score)
Saturday	
Sunday	Develop comparison table format for final results.

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

preparing evaluation Script.

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

DATE

PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	07-01
Week Ending Date	07-07

DAY	ACTIVITY
Monday	Review mAP, Precision, recall, and F1-Score results across all models.
Tuesday	
Wednesday	Compare model performance using bar graphs and tables
Thursday	
Friday	Document strenght and weaknesses of each model.
Saturday	
Sunday	Identify performance variation by traffic sign type.

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

Comparing performance variation
of each model.

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

2005/07/08
DATE

[Signature]
PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	07-08
Week Ending Date	07-14

DAY	ACTIVITY
Monday	
Tuesday	Deciding which model is best suited based on performance and speed.
Wednesday	
Thursday	Justify best model selection in terms of real world usability.
Friday	
Saturday	
Sunday	Document trade offs between accuracy and speed.

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

Selecting best model based
on performance.

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

DATE

PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	07-15
Week Ending Date	07-21

DAY	ACTIVITY
Monday	Document model selection rationale in log.
Tuesday	
Wednesday	Drafting the model selection of the thesis report.
Thursday	
Friday	
Saturday	Define user interface requirements for the system.
Sunday	

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

User interface requirement defining.

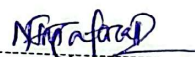
Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

2025/07/22
DATE


PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	07-22
Week Ending Date	07-28

DAY	ACTIVITY
Monday	choose interface development Library/frame (Streamlit)
Tuesday	
Wednesday	Setup development environment (IDE, Python packages)
Thursday	
Friday	Design Simple UI as a testing.
Saturday	
Sunday	

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

Setting up development environment for interface development.

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

DATE

PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	07-29
Week Ending Date	08-04

DAY	ACTIVITY
Monday	
Tuesday	Preparing model loading script for real-time inference.
Wednesday	
Thursday	Testing sample inter functionality with sample input.
Friday	
Saturday	
Sunday	Connect the model with interface.

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

Connecting model with Interface.

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

2005/08/05

DATE



PROJECT SUPERVISOR'S SIGNATURE

Project Progress Daily Sheet

Week Start Date	08-05
Week Ending Date	08-11

DAY	ACTIVITY
Monday	Crafting a complete interface with functionality.
Tuesday	Testing realtime responsiveness of model predictions.
Wednesday	
Thursday	Document all the research work in thesis report
Friday	thesis report over update.
Saturday	preparing final presentation
Sunday	

Details of Work Carried out, Problem encountered how it is solved and suggestions given by the supervisor (Sketches and diagrams should be given where ever it is applicable).

Major Tasks Completed:

final presentation preparation.

Problems Occurred:

Suggestions Made by the Supervisor:

Decisions Taken:

REMARKS AND CERTIFICATION BY THE LECTURER IN CHARGE/SUPERVISOR

2004/08/14

DATE



PROJECT SUPERVISOR'S SIGNATURE